

Rocky planet

A planet composed mainly of rock. The four rocky planets in the Solar System are Mercury, Venus, Earth, and Mars. See also *gas giant*

S**Satellite**

A natural satellite is an astronomical body that orbits a planet, otherwise known as a moon. An artificial satellite is an object deliberately put in orbit around Earth or another planet.

Seyfert galaxy

A spiral galaxy with an exceptionally bright central region. Seyfert galaxies are believed to be similar to quasars, although less powerful and found closer to our own galaxy. See also *active galaxy*

Singularity

A point of infinite density into which matter has been compressed by gravity, and a point at which the known laws of physics break down. Theory implies that a singularity exists at the center of a black hole. See also *black hole*

Solar flare

A violent release of huge amounts of energy from a localized region on the surface of the Sun.

Solar System

The Sun together with the eight planets, smaller bodies (dwarf planets, moons, asteroids, comets, trans-Neptunian objects), dust, and gas that orbit the Sun.

Solar wind

A constant stream of fast-moving particles that escapes from the Sun and flows outward through the Solar System.

Spacetime

The combination of the three dimensions of space (length, breadth, height) and the single time dimension. See also *relativity*

Spiral galaxy

A galaxy that consists of a central concentration of stars surrounded by a flattened disk of stars, gas, and dust, within which the major visible features are clumped together into spiral arms. See also *barred spiral galaxy*

Star

A huge sphere of glowing plasma that generates energy by means of nuclear reactions at its center. See also *fusion*, *plasma*

Star cluster

A group of stars bound together by gravity. See also *globular cluster*, *open cluster*

Subgiant star

A star that is significantly more luminous than a main-sequence star of the same surface temperature and color.

Sunspot

A region of intense magnetic activity in the Sun's photosphere. Sunspots appear dark in images because their temperature is lower than the rest of the photosphere. See also *photosphere*

Super-Earth

An extra-solar planet whose mass is greater than Earth's but less than planets such as Uranus and Neptune. See also *extra-solar planet*.

Supergiant star

An exceptionally luminous star with a very large diameter.

Supernova

A violent explosion of a massive star, during which it expels most of its matter, and its brightness increases hugely for a short time. A different kind of supernova happens when a white dwarf explodes after attracting material from a neighboring star.

T**Trans-Neptunian object**

A Solar System body orbiting the Sun beyond the orbit of Neptune.

U**Ultraviolet radiation**

Electromagnetic radiation with wavelengths shorter than visible light but longer than X-rays. See also *electromagnetic radiation*

Universe

The totality of matter, energy, and space that came into being as a result of the Big Bang.

V**Variable star**

A star that varies in brightness. A pulsating variable star physically expands and contracts in a regular rhythm, varying in brightness as it does so. An eruptive variable star brightens and fades abruptly. See also *Cepheid variable*, *Mira variable*, *eclipsing binary*

W**Wavelength**

The distance between two successive crests in a wave motion.

White dwarf star

A small, but very hot and dense, glowing body that remains after a star of similar mass to our Sun dies and sheds its outer layers into space.

Wolf-Rayet star

A massive, very hot star from which gas is escaping at an exceptionally rapid rate.

X**X-ray**

Electromagnetic radiation with wavelengths shorter than ultraviolet radiation but longer than gamma rays. See also *electromagnetic radiation*

Z**Zenith**

The point in the sky directly above an observer.

Zodiac

An imaginary band around the celestial sphere, through which the Sun, Moon, and planets appear to travel. It represents the plane of the Solar System as seen from Earth. See also *ecliptic*